

Computational design of biopolymer aerogels and predictive modelling of their nanostructure and mechanical behaviour

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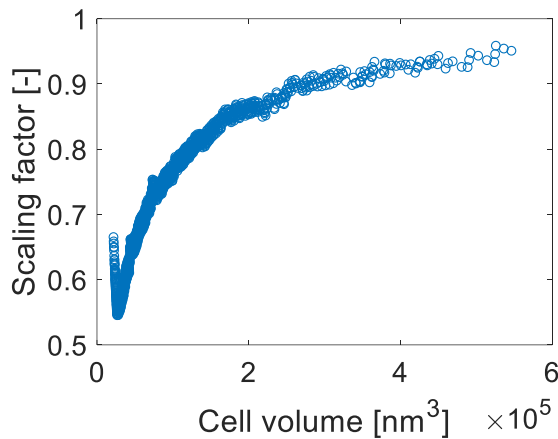


Figure S1: Scaling factor to obtain the Voronoi cell volume distribution

Additional details about Table 1. The envelope density increased from 0.07 to 0.15 g cm⁻³ with the increase of κ -carrageenan concentration from 1 to 3 wt.%, respectively. Brunauer-Emmett-Teller (BET) specific surface area was in the range between 221-228 m² g⁻¹ showing no significant influence while increasing the concentration from 1 to 3 wt.%. Recently, this synthetic procedure was reproduced by Agostinho *et al.* (Materials Chemistry and Physics 253: 123290 (2020)), and the data of specific surface area and total volume shrinkage was in good agreement with our previous work (Ganesan and Ratke, Soft Matter 10:3218 (2014)). The mesopore volume of aerogels was in the range between 1.37 and 1.97 cm³ g⁻¹. The lower the concentration of κ -carrageenan, the lower was the porosity of the aerogels. The skeletal density (ρ_s) of κ -carrageenan was 1.72 g cm⁻³ as obtained from helium pycnometer. Using this value in Eq. (3), the fibre thickness was estimated, and it was in the range between 10.2 and 10.55 nm. It indicated that the fibre thickness was not significantly influenced by increasing the concentration of κ -carrageenan.

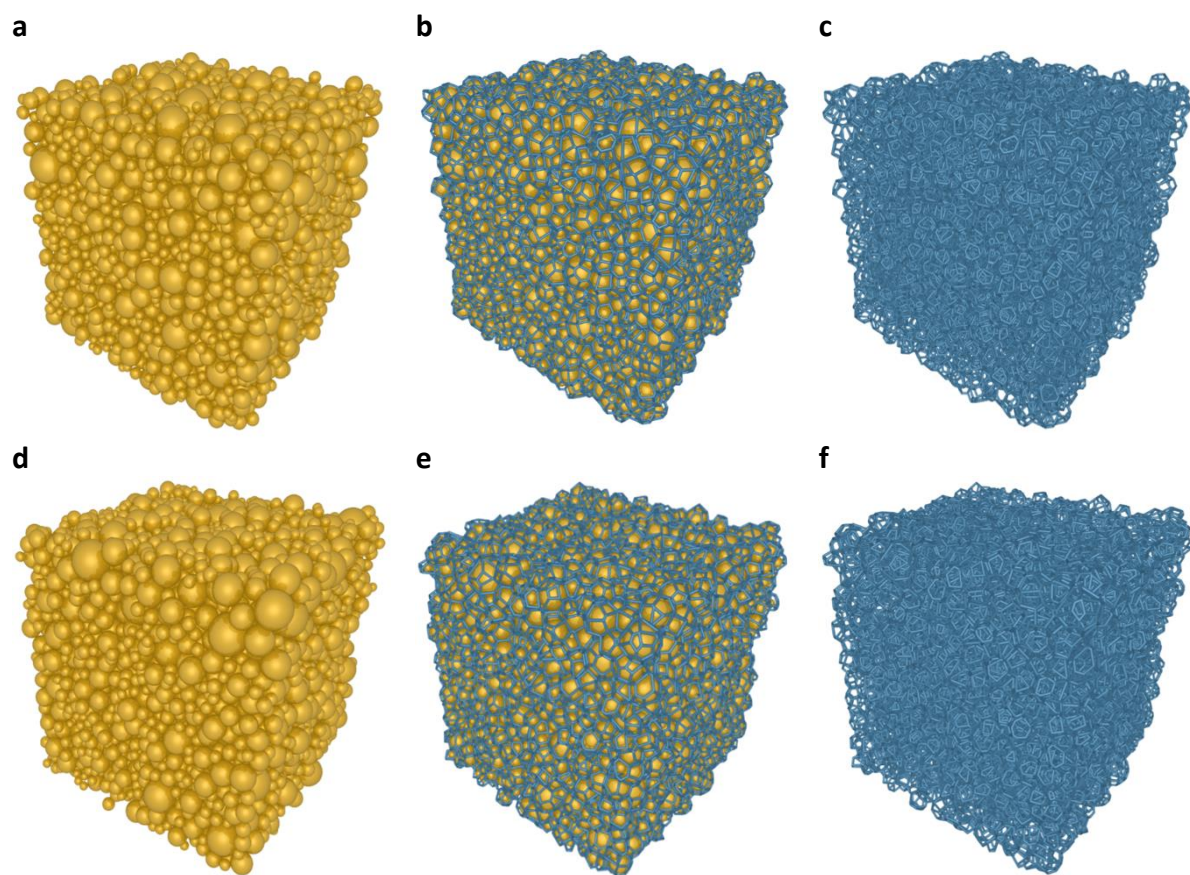


Figure S2: **a, d** Sphere-packed simulation box, **b, e** the corresponding LVT diagram over the sphere-packed box and **c, f** the computationally designed nanostructured aerogel network corresponding to 2 wt.% and 3 wt.% κ -carrageenan aerogels, respectively.

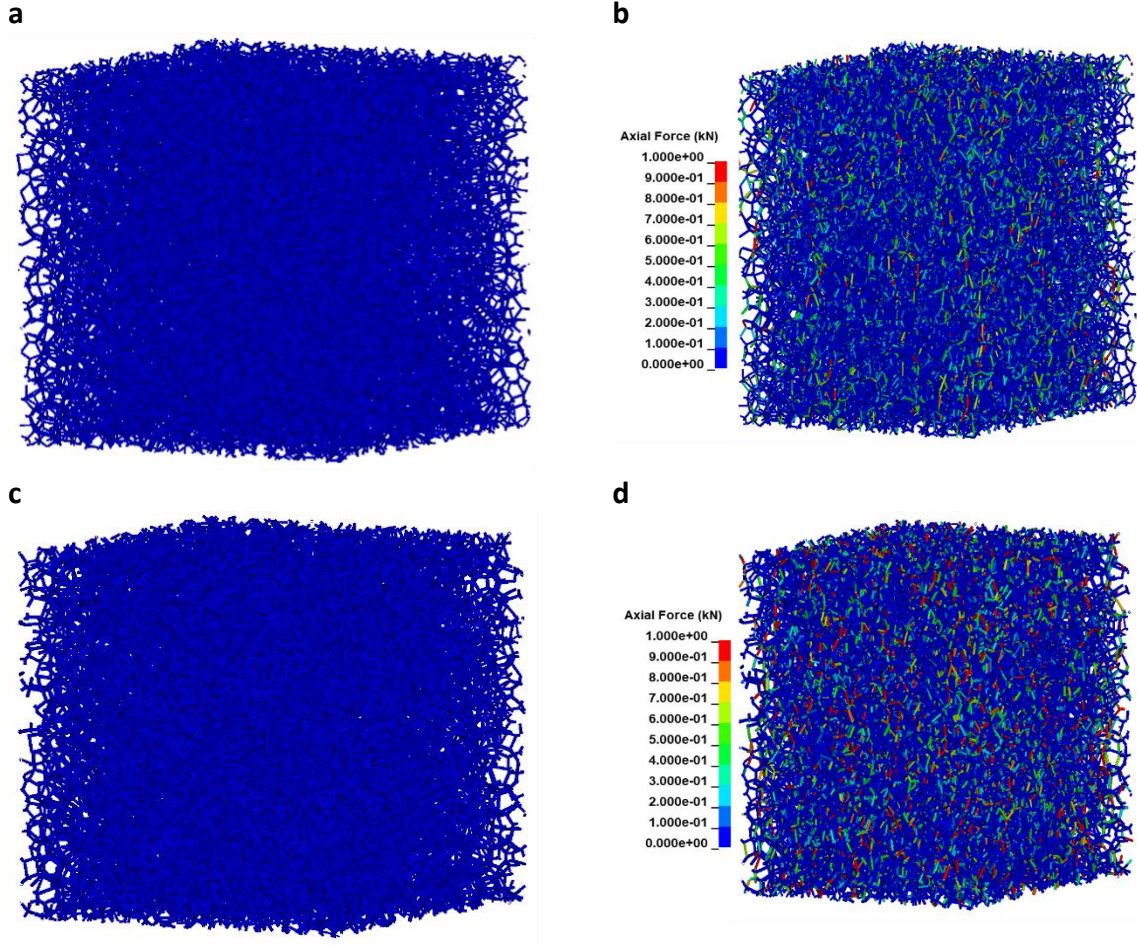


Figure S3: The RVE in the reference and deformed configuration of **a-b** 2 wt.% and **c-d** 3 wt.% κ -carrageenan aerogels, respectively.

Algorithm 1 Calculate fibre diameter

- 1: Calculate the volume of each Voronoi cell: V_j
 - 2: Calculate the solid fraction of each Voronoi cell: $\bar{V}_j = \phi_s V_j$
 - 3: Initialise the edge volume to 0 for all edges
 - 4:
 - 5: **for** each Voronoi cell **do**
 - 6: Calculate the set of edges adjacent to the cell: S
 - 7: Calculate the sum of the edge lengths in S : L_S
 - 8: Calculate weighting factors for each edge in S based on its length/ L_S
 - 9: Calculate the volume share for each edge in S using $\bar{V}_j$ and the weighting factors
 - 10: Add the determined edge volume share to the edge volumes
 - 11:
 - 12: **for** each edge **do**
 - 13: Calculate the required fibre diameter based on the edge volume
 - 14:
 - 15: **return** fibre diameter for each edge
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Figure S4: Algorithm illustrating the step-by-step procedure to obtain individual fibre diameters in the computationally generated Voronoi-based aerogel microstructure.